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Flame–vortex interaction driven combustion dynamics in a backward-facing step combustor

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ABSTRACT

The combustion dynamics of propane–hydrogen mixtures are investigated in an atmospheric pressure, lean, premixed backward-facing step combustor. We systematically vary the equivalence ratio, inlet temperature and fuel composition to determine the stability map of the combustor. Simultaneous pressure, velocity, heat release rate and equivalence ratio measurements and high-speed video from the experiments are used to identify and characterize several distinct operating modes. When fuel is injected far upstream from the step, the equivalence ratio entering the flame is temporally and spatially uniform, and the combustion dynamics are governed only by flame–vortex interactions. Four distinct dynamic regimes are observed depending on the operating parameters. At high but lean equivalence ratios, the flame is unstable and oscillates strongly as it is wrapped around the large unsteady wake vortex. At intermediate equivalence ratios, weakly oscillating quasi-stable flames are observed. Near the lean blowout limit, long stable flames extending from the corner of the step are formed. At atmospheric inlet temperature, the unstable mode resonates at the 1/4 wavemode of the combustor. As the inlet temperature is increased, the 5/4 wavemode of the combustor is excited at high but lean equivalence ratios, forming the high-frequency unstable flames. Higher hydrogen concentration in the fuel and higher inlet temperatures reduce the equivalence ratios at which the transitions between regimes are observed. We plot combustion dynamics maps or the response curves, that is the overall sound pressure level as a function of the equivalence ratio, for different operating conditions. We demonstrate that numerical results of strained premixed flames can be used to collapse the response curves describing the transitions among the dynamic modes onto a function of the heat release rate parameter alone, rather than a function dependent on the equivalence ratio, inlet temperature and fuel composition separately. We formulate a theory for predicting the critical values of the heat release parameter at which quasi-stable to unstable and unstable to high-frequency unstable modes take place.

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1. Introduction

Continuous combustion systems common in propulsion and energy applications are susceptible to thermoacoustic instability. The instability may occur under lean burn conditions, where most emissions and efficiency benefits are achieved, or near stoichiometry, where high power density is the objective. As a result of the resonant feedback interactions between the driving heat release mechanisms and the acoustic modes, these combustors are known to exhibit significant pressure and flow oscillations, which may cause flame extinction, structural vibration, flame flashback and even structural damage [1,2]. Several mechanisms are known to promote such interactions: flame–acoustic wave interactions, flame–vortex interactions, equivalence ratio fluctuations, flame–wall interactions and the effect of unsteady stretch rate, all of which may be present individually or simultaneously [3–5].

In this paper we explore the mechanism of flame–vortex interactions, which is one of the most significant instability mechanisms in large-scale gas turbine combustors [6,7]. As a result of the unsteady interaction between the flame and the vortex (vortices) formed in the wake of the recirculation zone(s), the heat release rate oscillates significantly. When the periodic heat release supported by the flame–vortex interaction couples positively with the acoustic field, self-sustained acoustic oscillations are formed [8–16].

Recently, in order to broaden the operating range of combustors, hydrogen enriched hydrocarbons have been used as fuels in LPP combustors [17–20]. Hydrogen enrichment allows operation at lower flame temperature, which helps reduce NO_x emissions and the equivalence ratio at the lean blowout. Observed combustion dynamics and flame structures are also impacted by hydrogen enrichment as a result of its impact on the reaction kinetics. Previous

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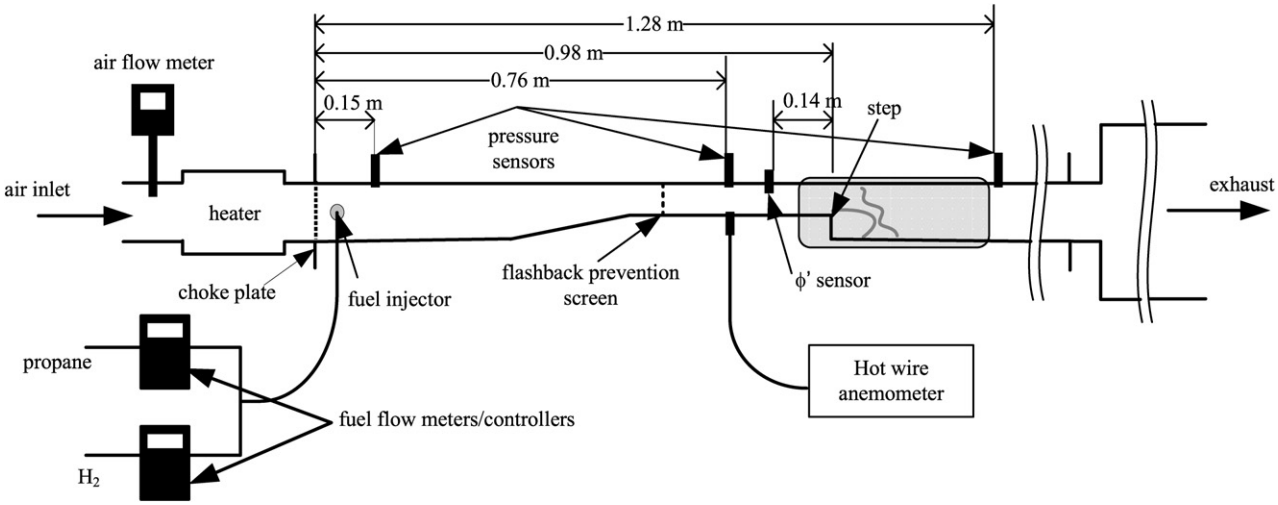

Fig. 1. Schematic diagram of the backward-facing step combustor.

research has shown that in the case of unstrained methane–air flames, hydrogen addition has a relatively small impact on the laminar burning velocity and the lean blowout limit. For methane–hydrogen mixtures with 10% fuel volume of H_2, the increase in burning velocity is typically 5%, over a range of equivalence ratios [21]. In contrast, hydrogen enrichment has been shown to substantially increase the burning velocity and to inhibit extinction in turbulent and strained flame environments [22–24]. Analytical studies have described the impact of stretch on premixed flames, typically using asymptotic analysis and simplified models of flame structure [25–27]. These studies yield simple expressions for the variation of the burning velocity and flame temperature from their unstretched values, emphasizing the interaction of stretch with preferential diffusion (e.g. non-unity Lewis number) effects. The impact of inlet temperature on thermoacoustic instabilities, flame structure and blowout limits have also been investigated and noticeable shifts in regions of unstable combustion have been observed [28,29].

In this paper, we carry out a parametric study by varying the equivalence ratio, Reynolds number, inlet temperature and the fuel composition to examine the flame–vortex interaction driven combustion dynamics in an atmospheric backward-facing step combustor. Extensive instrumentation of the combustor allows monitoring the temporal variations in pressure, heat release, and flow velocity, as well as the temporal and spatial variations in equivalence ratio. We fired the combustor at Reynolds numbers of 6500 and 8500 based on the step height using pure propane or propane enriched with hydrogen by adding 30% by volume (2.0% by mass; 4.8% by LHV) or 50% by volume (4.4% by mass; 10.6% by LHV) hydrogen. The temperature of the inlet mixture is varied from ambient (300 K) to 600 K, in 100 K increments. At each fuel composition, Reynolds number and the inlet temperature, we varied the equivalence ratio of the fuel–air mixture from near the lean blowout limit to a value approaching the flashback limit near stoichiometry. We conducted the experiments with the fuel injector located 93 cm upstream of the step, which ensures that the equivalence ratio reaching the combustion zone is temporally and spatially uniform.

We demonstrate the presence of different regimes in which the flame is impacted by different flow structures depending on the operating parameters. The response curves of the combustor—the pressure amplitude as function of the equivalence ratio—are presented for a range of fuel compositions and inlet temperatures. We use numerical results to collapse the response curves, showing that the transitions between the combustor dynamic modes are correlated with changes in a heat release parameter which incorporates the effect of strained flame consumption speed and volumetric expansion. We show that the complex interactions between the hydrodynamics and the flames under different operating conditions can approximately be captured using the heat release parameter alone. Formulating a simple theory, we determine the critical values of this parameter at which transitions between different dynamic operating modes take place.

The article is structured as follows. In Section 2, we describe the experimental setup and the diagnostic tools. In Section 3, we report the results describing the stability map of the combustor, the combustor acoustics, the high speed flame images and the derived theory using strained flame consumption speeds. We use acoustic analysis to determine the unstable modes analytically, and explain the observations. We use strained flame calculations to find a "reducing" parameter that can collapse the measurements. In Section 4, the article ends with conclusions.

2. Combustor description

Fig. 1 shows a schematic diagram of the backward-facing step combustor. The combustor consists of a rectangular stainless steel duct with a cross section 40 mm high and 160 mm wide. The air inlet to the combustor is choked. At a location 0.45 m downstream from the choke plate, a 0.15 m long ramp contracts the channel height from 40 mm to 20 mm followed by a 0.4 m long constant area section that ends with a sudden expansion back to 40 mm. The step height is 20 mm. The overall length of the combustor is 5.0 m. A circular exhaust pipe comprises the last 3.0 m of the combustor with a cross sectional area approximately four times that of the rectangular section. The exhaust exits to a trench with a large cross sectional area. The combustor is equipped with quartz viewing windows. An air compressor supplies air up to 110 g/s at 883 kPa. A pair of Sierra C100M mass flow controllers allow arbitrary propane/hydrogen mixtures at maximum flow rates of 2.36 g/s for propane and 0.30 g/s for hydrogen. The uncertainty of the flow rates is ±1% of the full scale. Fuel is injected through several spanwise holes in a manifold located 93 cm upstream of the step. Images of the flame are captured at 2000 frames per second using a Phantom v7.1 high-speed camera. Pressure measurements are obtained using Kulite MIC-093 high intensity microphones designed for laboratory investigations. Flow velocity is measured using TSI IFA300 hot wire anemometer. An optical bandpass filter centered at 430 nm is placed in front of a Hamamatsu H9306-02 photosensor module to measure the CH* chemiluminescence emitted by the flame, which is proportional to the instantaneous

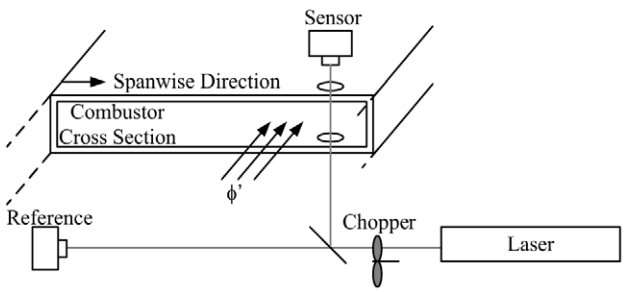

Fig. 2. Schematic diagram showing the temporal equivalence ratio measurement setup.

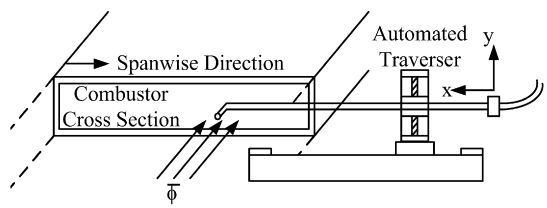

Fig. 3. Schematic diagram showing the spatial equivalence ratio measurement setup.

heat release rate [30,31]. More details on CH* chemiluminescence measurement technique can be found in Ref. [16]. Temporal equivalence ratio variations are measured 14 cm upstream of the step using a Hamamatsu P4245 photodiode to detect absorption of a HeNe laser by propane at a wavelength of 3.39 μm. This laser is modulated using a Scitec Instruments 360 OEM optical chopper in order to distinguish between the light emitted by the flame when it flashes back and the laser signal. When the fuel passes through the laser beam, it absorbs some of the laser light, reducing the photodiode signal. The intensity of the light can be related to the fuel concentration using the Beer–Lambert law, as described by Lee et al. in Ref. [32]. Fig. 2 shows the schematic diagram of the temporal equivalence ratio measurement setup. The spatial concentration/equivalence ratio measurements are performed under non-reacting flow conditions by injecting CO_2 into the combustor as a fuel surrogate and taking point measurements of its concentration using a California Analytical Instruments ZRH CO/CO_2 analyzer. The gas probe automatically traverses the combustor cross section near the location of the step. The spatial concentration measurement setup is shown schematically in Fig. 3. The inlet mixture temperature is set using an Osram Sylvania 18 kW in-line electric heater with on/off temperature controller. All data are acquired using a National Instruments PCIe-6259 data acquisition board and the Matlab Data Acquisition Toolbox. A custom Matlab code is used to store the data and control the experiment.

3. Results

Temporal oscillations of the equivalence ratio, and non-uniformities in the equivalence ratio distribution entering the flame zone can significantly impact the combustion dynamics by impacting the local burning velocity along the flame front as well as the flame structure. Here, since we are interested in investigating only the flame–vortex interaction driven combustion dynamics, we ensure that the equivalence ratio entering the combustion zone is steady, i.e. no temporal variations of the equivalence ratio is present at the combustion zone. Moreover, in order to achieve premixed combustion, the spatial equivalence ratio distribution entering the flame zone should be uniform.

In Fig. 4 we show the temporal variation of the equivalence ratio measured 14 cm upstream of the step when the combustor is fired at an equivalence ratio of 0.80 and a Reynolds number

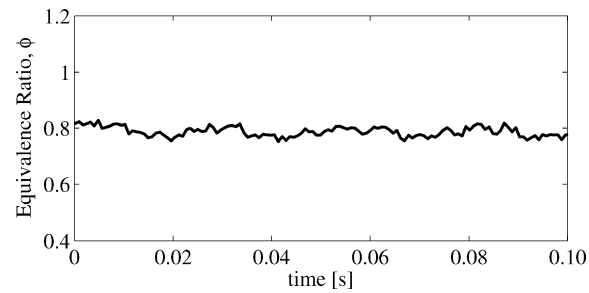

Fig. 4. Temporal equivalence ratio measurements 14 cm upstream of the step at Re = 8500, $\phi = 0.80$ while injecting the fuel 93 cm upstream of the step.

of 8500, without hydrogen enrichment, when the fuel injector is located 93 cm upstream of the step. At this location of the fuel injector no temporal variations of the equivalence ratio are observed as a result of two factors: (i) the amplitude of equivalence ratio oscillations established at the fuel injector location are small, since the injector is located near the choke plate where velocity oscillations are small; and (ii) the convective time scale, $\tau = L/\bar{U}$ is long so that there is enough time for the oscillations to be damped by turbulent mixing before reaching the combustion zone, where L is the distance between the fuel injector and the step and $\bar{U}$ is the average flow velocity [4,33].

Next, we measure the spatial distribution of the equivalence ratio keeping the fuel injector at the same location, under steady, non-reactive conditions using the technique described in the previous section. The measurements are performed at a cross section immediately downstream of the step, without hydrogen enrichment, at Re = 8500 and mean equivalence ratio of 0.70. The maximum variation of the measured equivalence ratio distribution along the cross section is less than 1%. This indicates uniform spatial equivalence ratio distribution, ensuring premixed combustion.

In the rest of the paper, in order to investigate flame–vortex interaction driven premixed combustion dynamics, we introduce the fuel 93 cm upstream of the step as suggested by the above results. We conduct a series of experiments in order to: (i) determine the stability map; (ii) investigate the resonant acoustic modes; (iii) distinguish between the flame–vortex interaction dynamics observed under distinct operating modes. Finally, we formulate a theory supported by numerical simulations of one-dimensional, strained flames to predict the transitions between different operating modes of the combustor.

3.1. Stability map

We systematically vary the inlet temperature, fuel composition and the Reynolds number, and measure the response curves of the combustor while varying the equivalence ratio between the lean blowout limit and near stoichiometry. We report the measured pressure fluctuations in terms of the overall sound pressure level (OASPL) using the pressure sensor located 1.28 m downstream of the choke plate, which is close to the location of the flame. The OASPL in dB is defined as:

$$\mathrm{OASPL} = 10\log_{10}\overline{\left[\frac{p(t)-\overline{p(t)}}{p_o}\right]^2} \tag{1}$$

where overbars indicate average values, $p(t)$ is the pressure measured in an interval $t_1 < t < t_2$ and $p_o = 2\times10^{-5}$ Pa.

While performing the tests, we observed that the combustor dynamics were strongly influenced by the history of the operating mode and the conditions. In order to illustrate this history dependence, we plot the OASPL as a function of equivalence ratio for propane flames at Reynolds numbers of 6500 and 8500 and the inlet temperature of 300 K in Fig. 5. At each Reynolds number,

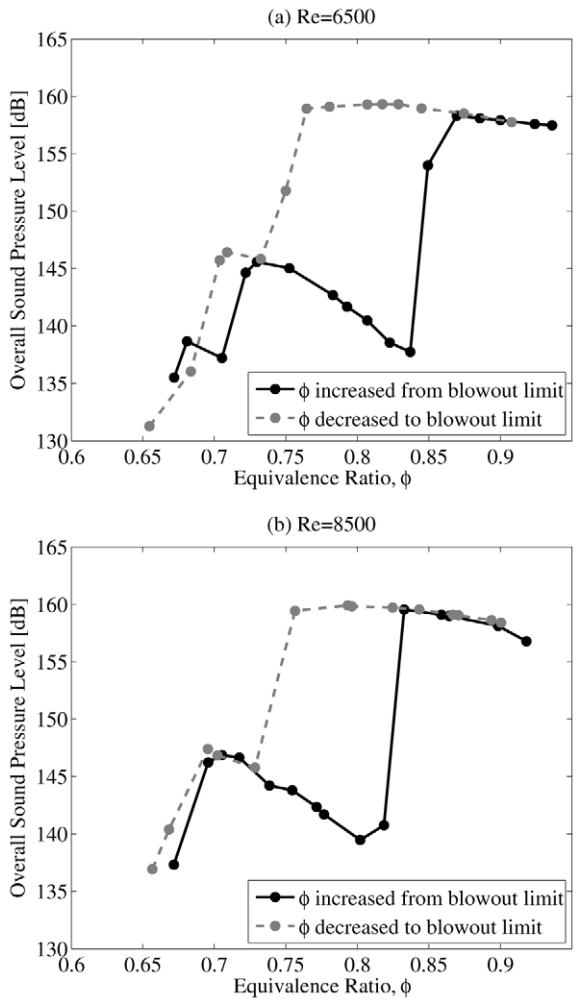

Fig. 5. OASPL as a function of equivalence ratio without hydrogen enrichment at an inlet temperature of 300 K. The equivalence ratio is either increased from near the blowout limit toward stoichiometry, or decreased from near stoichiometry toward the lean blowout limit for (a) Re = 6500; (b) Re = 8500.

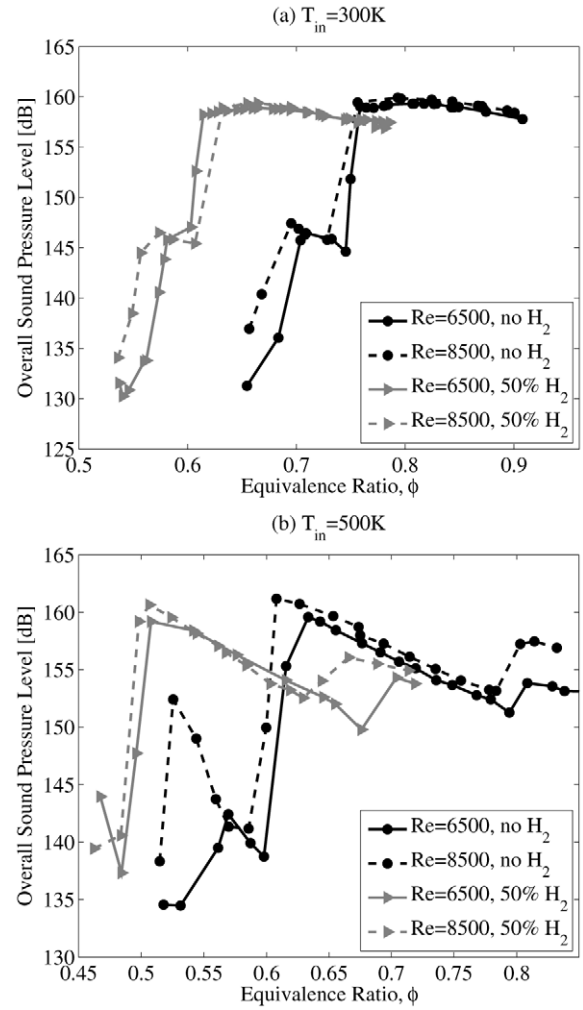

Fig. 6. OASPL as a function of equivalence ratio using pure propane and propane enriched with 50% by volume hydrogen at Re = 6500 and Re = 8500 at an inlet temperature of (a) $T_{in} = 300$ K; (b) $T_{in} = 500$ K.

the equivalence ratio is either increased from the lean blowout limit toward stoichiometry, or decreased from near stoichiometry toward the lean blowout limit. When we increase the equivalence ratio from the lean blowout limit, the transitions in the response curves shown in Fig. 5 shift toward higher equivalence ratios at both Reynolds numbers. The dependence of the combustor dynamics on the history, i.e. the presence of hysteresis, is attributed to the highly non-linear nature of the system. In the rest of the paper, the data are obtained while decreasing the equivalence ratio from near stoichiometry toward the lean blowout limit.

In Fig. 6, we plot the OASPL as a function of the equivalence ratio using pure propane or propane enriched with 50% by volume (4.4% by mass; 10.6% by LHV) hydrogen at Reynolds numbers of 6500 and 8500, and inlet temperatures of 300 K and 500 K. When the inlet temperature is 300 K, we observed three distinct operating modes corresponding to sound pressure levels of 158–160 dB, 145–150 dB and 130–135 dB. The combustor operating modes corresponding to the 158–160 dB bands are referred to as *unstable*, 145–150 dB bands are referred to as *quasi-stable* and OASPL values between 130–135 dB close to the lean blowout are referred to as *stable*. The origin of each characterization will become apparent when we depict the flame images during the corresponding operating modes. When the inlet temperature is raised to 500 K, the OASPL in the unstable region decreases slowly with increasing equivalence ratio and suddenly jumps to much higher values as the equivalence ratio exceeds a certain threshold (0.80 without hydrogen enrichment at both Reynolds numbers). This is the *high-frequency unstable* mode, where a higher acoustic mode of the combustor is excited. The impact of the Reynolds number on the stability map is minor. The quasi-stable mode is wider when the Reynolds number is 8500, especially at inlet temperature of 500 K. Enriching propane with hydrogen or increasing the temperature of the inlet mixture lower the lean blowout limit and shift the observed transitions between operating modes to lower equivalence ratios.

We investigate the results in Fig. 6 in more detail by showing the combustor's frequency response in Figs. 7 and 8 for the cases corresponding to Figs. 6(a) and 6(b), respectively. These figures show the amplitude of pressure oscillations as a function of frequency for the ranges of equivalence ratios in Fig. 6. High amplitudes correspond to darker regions.

Fig. 7(a) shows the frequency response for Reynolds number of 6500, inlet temperature of 300 K, without hydrogen enrichment. As the equivalence ratio is dropped from 0.90 to 0.70, the acoustic mode around 40 Hz is excited and the combustor operates in the unstable mode. As the equivalence ratio is reduced just below 0.70, there is a slight increase in the resonant frequency, and a transition to the quasi-stable operating mode. The resonant frequency drops slowly, as the equivalence ratio is decreased further until the lean blowout limit. Similar acoustic response is observed in Fig. 7(b),

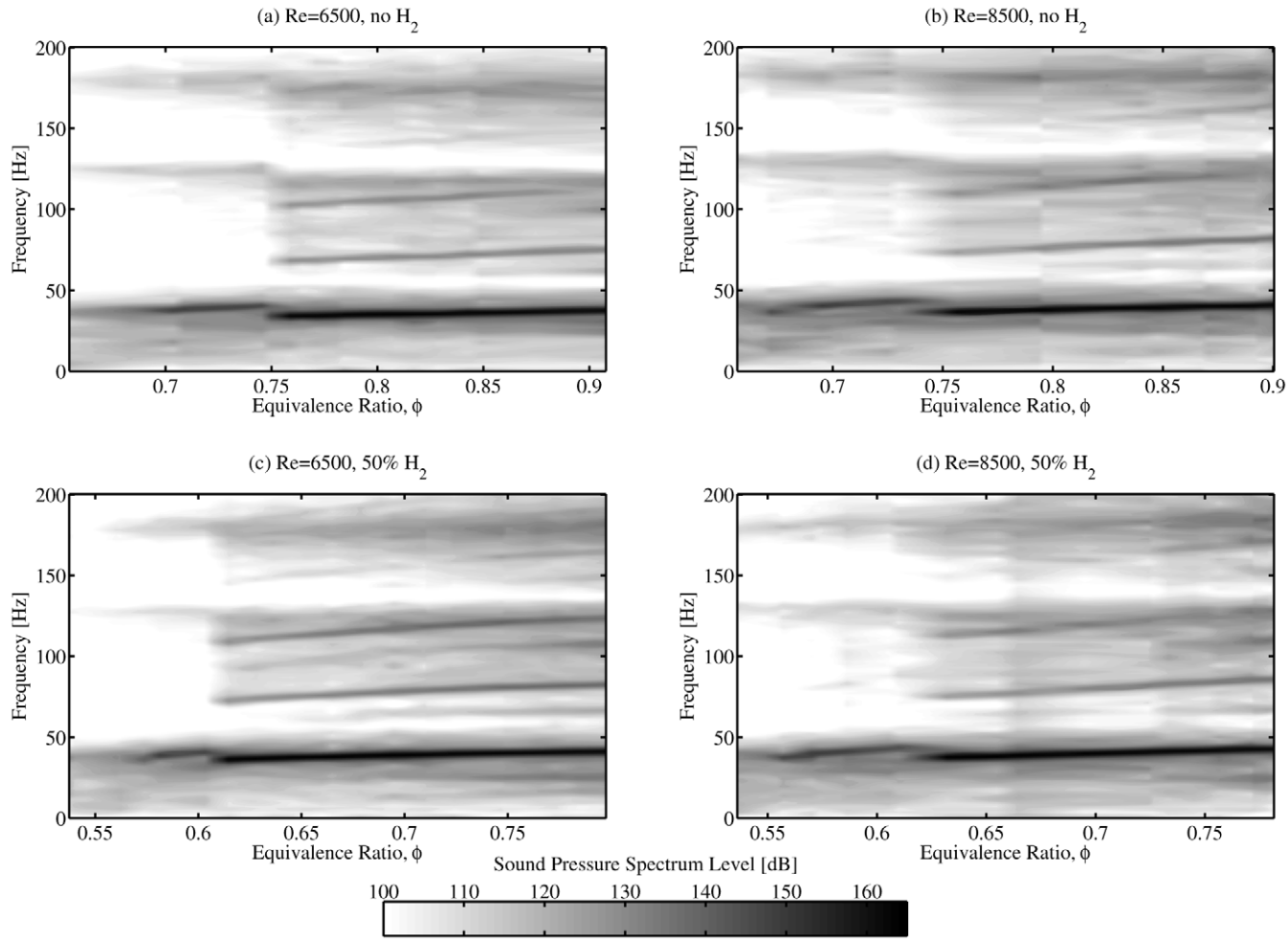

Fig. 7. Sound pressure spectrum level maps as a function of equivalence ratio at inlet temperature of 300 K for (a) Re = 6500, no H_2 enrichment; (b) Re = 8500, no H_2 enrichment; (c) Re = 6500, 50% by volume H_2 enrichment; (d) Re = 8500, 50% by volume H_2 enrichment. Dark bands indicate high amplitude frequency components.

which shows the frequency response for Reynolds number of 8500, inlet temperature of 300 K, without hydrogen enrichment. However, at this Reynolds number, the transition to the quasi-stable mode occurs around the equivalence ratio of 0.73, which is slightly higher than that in the Re = 6500 case. When 50% by volume hydrogen is added to propane, the transition to quasi-stable mode occurs at an equivalence ratio of 0.62 at both Reynolds numbers, as shown in Figs. 7(c) and 7(d). When the inlet temperature is 300 K, only the resonant mode around 40 Hz is dominant for the unstable and the quasi-stable operating modes of the combustor.

When the inlet temperature is raised to 500 K, there are significant changes in the acoustic response of the combustor as shown in Fig. 8. Without hydrogen addition, at both Reynolds numbers, as the equivalence ratio drops below 0.85, the combustor resonates around 130 Hz, until the equivalence ratio reaches 0.80, as shown in Figs. 8(a) and 8(b). These operating bands correspond to OASPLs of 153 dB and 158 dB for Re = 6500 and Re = 8500, respectively, in Fig. 6(b), and are referred to as the high-frequency unstable mode. As the equivalence ratio is reduced further, the 40 Hz band is excited until blowout occurs. However, near $\phi = 0.60$, at both Reynolds numbers, the OASPL drops significantly, corresponding to the transition from an unstable to a quasi-stable band. The OASPL increases until the equivalence ratio reaches 0.57 for Re = 6500 and 0.52 for Re = 8500. When 50% by volume hydrogen is added to propane, as the equivalence ratio is reduced from 0.72, the combustor resonates at 130 Hz until the OASPL reaches its local minimum around equivalence ratios of 0.68 for Re = 6500 and 0.65 for Re = 8500, as shown in Fig. 6(b). As the equivalence ratio is reduced further, the 40 Hz band is excited until the combustor shifts from unstable to quasi-stable operating mode around equivalence ratio of 0.49 at both Reynolds numbers.

So far we have demonstrated that changing the hydrogen content in the fuel and the inlet temperature affect the response curves significantly. Changing the Reynolds number has a minor impact. We investigate the impact of hydrogen enrichment and the inlet temperature further by extending the parameter set to include two additional inlet temperatures and one additional hydrogen concentration. In Fig. 9, we plot the OASPL as a function of the equivalence ratio at Reynolds number of 6500 for different fuel compositions and inlet temperatures. Fig. 9 shows that at all inlet temperatures, increasing the hydrogen fraction in the fuel shifts the response curves toward leaner equivalence ratios. Increasing the inlet temperature at constant fuel composition has the same impact on the response curves. With preheating, the OASPL drops as the equivalence ratio increases until the high-frequency unstable mode is excited, as shown in Figs. 9(c) and 9(d). At the local minimum, which is the transition point between the unstable and the high-frequency unstable modes, the combustor operates in the quasi-stable mode when the inlet temperature is 600 K. Moreover, the distinct quasi-stable band disappears at this inlet temperature. Preheating the inlet mixture allows the excitation of higher frequency modes, impacting the combustion dynamics significantly.

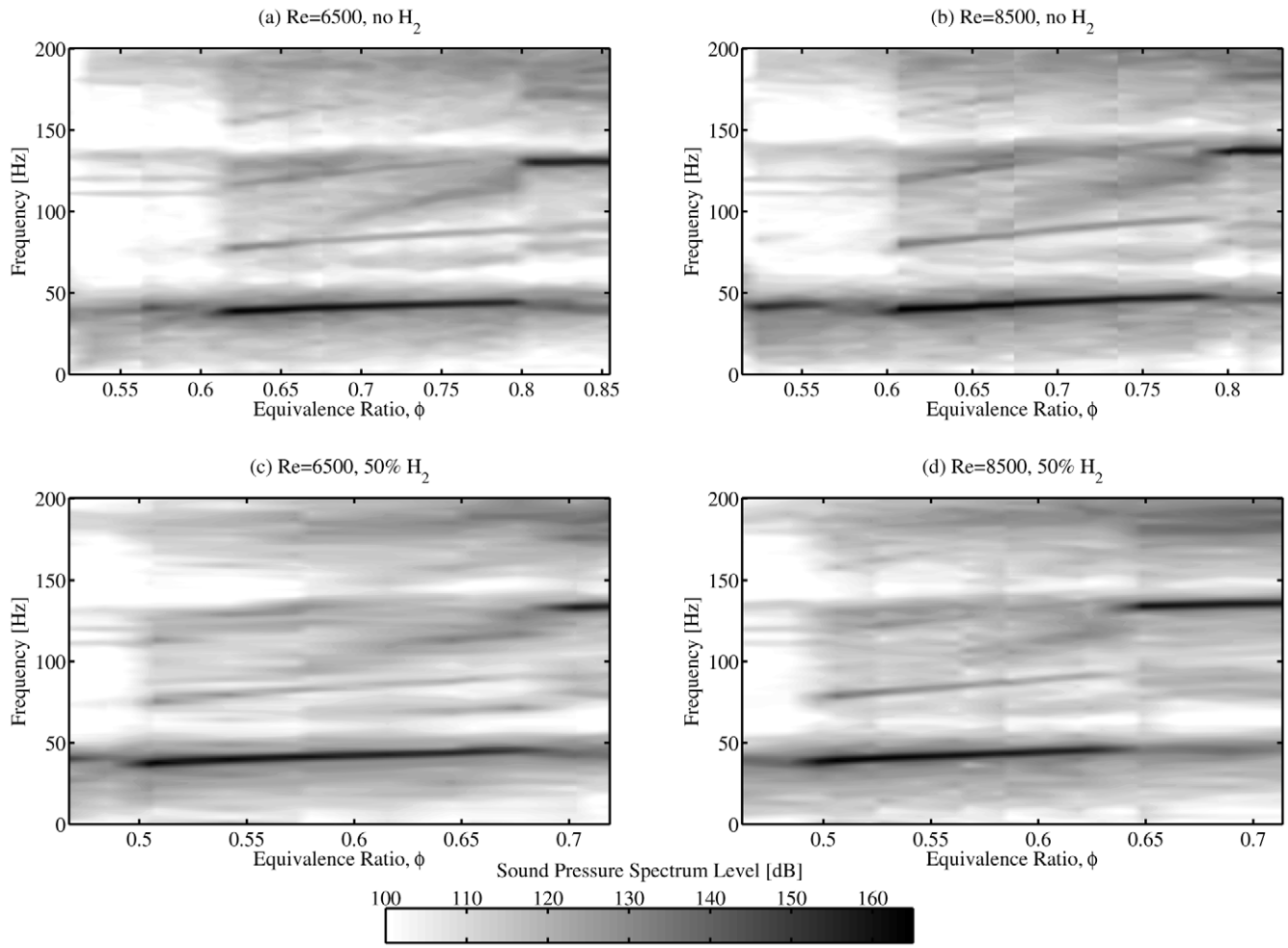

Fig. 8. Sound pressure spectrum level maps as a function of equivalence ratio at inlet temperature of 500 K for (a) Re = 6500, no H_2 enrichment; (b) Re = 8500, no H_2 enrichment; (c) Re = 6500, 50% by volume H_2 enrichment; (d) Re = 8500, 50% by volume H_2 enrichment. Dark bands indicate high amplitude frequency components.

3.2. Combustor acoustics

In the previous section, we showed that two different acoustic modes of the combustor, corresponding to frequencies of 40 Hz and 130 Hz, were excited under different operating conditions. In this section we investigate the combustor acoustics in detail to explain the origin of the observed resonant frequencies.

In Fig. 10, we plot the simultaneous pressure oscillations measured at four different locations along the length of the combustor corresponding to unstable, high-frequency unstable and quasi-stable modes. For the unstable and quasi-stable modes, the resonant frequency is close to 40 Hz and the pressure oscillations measured at different locations are nearly in phase. This shows that the resonant frequency may correspond to the 1/4 wavemode of the combustor. On the other hand, for the high-frequency unstable mode, a pressure node exists between the measurement locations of 0.15 m and 0.76 m (since the pressure measurements are out of phase); and another pressure node exists between measurement locations of 1.28 m and 3.82 m. The presence of two pressure nodes along the length of the combustor suggests that the resonant frequency around 130 Hz may correspond to the 5/4 wavemode. In Fig. 11 we plot the pressure modeshapes along the length of the combustor for unstable, high-frequency unstable and quasi-stable modes at the same operating conditions as in Fig. 10. The sign of the rms pressure amplitude is reversed whenever measurements at two consecutive locations are out of phase. Fig. 11 clearly shows that the quasi-stable and unstable modes correspond to the 1/4 wavemode, and the high-frequency unstable mode corresponds to the 5/4 wavemode of the combustor. The resonant frequency of 130 Hz is much smaller than what we expect for a 5/4 wavemode: 5×40 Hz = 200 Hz. Moreover, the pressure oscillations do not satisfy the open boundary condition at the end of the exhaust tube (at 5.0 m) as we expect, suggesting a longer acoustic length. It is also interesting to see that 3/4 wavemode is not excited.

In order to explain the observed resonant frequencies, we analytically modeled the combustor acoustics. The acoustics of the combustor are modeled by solving the linear one-dimensional wave equation while neglecting viscous effects. The Mach number of the flow in our combustor is very low (~0.02), therefore the effect of the mean flow is neglected. Fig. 12 shows a schematic diagram of the combustor. The combustor is divided into two regions by the combustion zone. For simplicity, we treat the combustion zone as a thin planar flame located at a fixed axial location. The effect of heat addition at the flame location is neglected for simplicity. The general solution of the homogeneous wave equation can be written in the following form [34]:

$$p'(x,t) = f(t - x/\bar{c}) + g(t + x/\bar{c}) \tag{2}$$

where the functions $f(t)$ and $g(t)$ are arbitrary. From the linearized one-dimensional momentum equation, the velocity oscillations are obtained in the following form:

$$u'(x,t) = \frac{1}{\bar{\rho}\bar{c}}\left[f(t - x/\bar{c}) - g(t + x/\bar{c})\right] \tag{3}$$

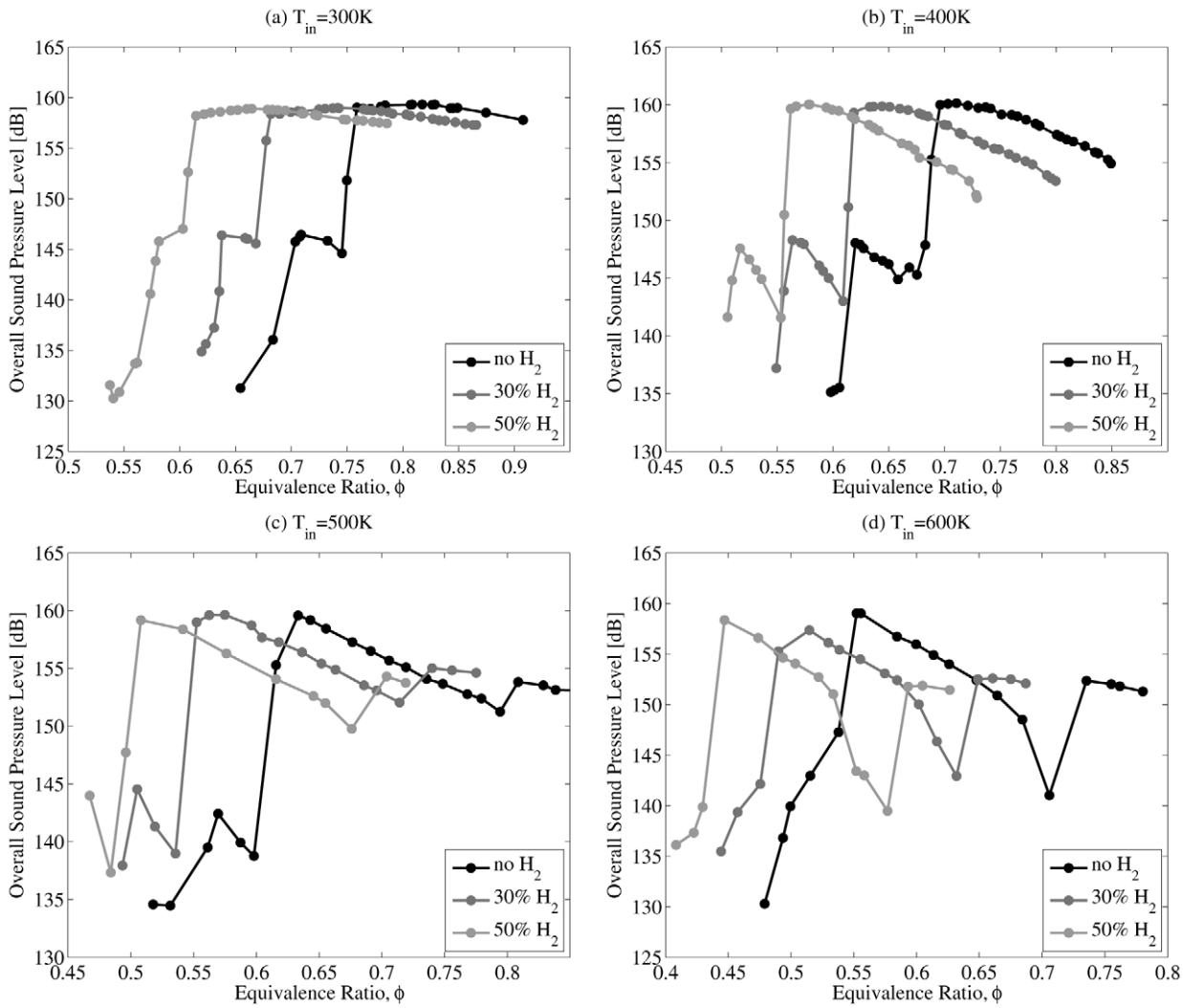

Fig. 9. OASPL as a function of equivalence ratio at different hydrogen concentration for (a) $T_{in} = 300$ K; (b) $T_{in} = 400$ K; (c) $T_{in} = 500$ K; (d) $T_{in} = 600$ K. Re = 6500.

where $\bar{\rho}$ and $\bar{c}$ are the average density and speed of sound, respectively. It is convenient to write $f(t) = Re[\hat{f}e^{i\omega t}]$ where the circumflex denotes a complex amplitude. Using this notation, we rewrite Eqs. (2) and (3) as follows:

$$\hat{p}(x) = \hat{f}e^{-i\omega x/\bar{c}} + \hat{g}e^{i\omega x/\bar{c}}, \tag{4}$$

$$\hat{u}(x) = \frac{1}{\bar{\rho}\bar{c}}\left[\hat{f}e^{-i\omega x/\bar{c}} - \hat{g}e^{i\omega x/\bar{c}}\right]. \tag{5}$$

In each section of the combustor, Eqs. (4) and (5) give the pressure and velocity mode shapes. The resonant frequencies follow from the application of the appropriate boundary conditions at the inlet and exit of the combustor and matching the solutions at the flame location.

The solution of the homogeneous wave equation in the upstream section satisfying the choked inlet boundary condition, $\hat{u}_1(0) = 0$ is:

$$\hat{p}_1(x) = A\cos(\omega/\bar{c}_1 x), \tag{6}$$

$$\hat{u}_1(x) = -\frac{iA}{\bar{\rho}_1\bar{c}_1}\sin(\omega/\bar{c}_1 x) \tag{7}$$

where A is an arbitrary constant. The solution of the homogeneous wave equation in the downstream section satisfying the open exit boundary condition, $\hat{p}_2(L_{tot}) = 0$ is:

$$\hat{p}_2(x) = B\sin\left[\omega/\bar{c}_2(L_{tot} - x)\right], \tag{8}$$

$$\hat{u}_2(x) = -\frac{iB}{\bar{\rho}_2\bar{c}_2}\cos\left[\omega/\bar{c}_2(L_{tot} - x)\right] \tag{9}$$

where B is an arbitrary constant. Matching the velocity and pressure solutions at the location of the flame, $x = L_f$, we obtain the following equation which is solved numerically to calculate the resonant frequencies:

$$\frac{\bar{\rho}_1\bar{c}_1}{\bar{\rho}_2\bar{c}_2} = \tan(\omega/\bar{c}_1 \cdot L_f)\tan\left[\omega/\bar{c}_2 \cdot (L_{tot} - L_f)\right]. \tag{10}$$

Fig. 13 shows the modeshapes and associated resonant frequencies corresponding to the 1/4, 3/4, 5/4 and 7/4 wavemodes of the combustor. The temperature of the reactants and products are taken as 300 K and 1880 K, respectively, corresponding to a pure propane flame with $\phi = 0.70$. The speed of sound is 342 m/s in the reactants and 830 m/s in the products. The density of the reactants is 1.194 kg/m^3 and the density of the products is 0.190 kg/m^3. We assume that the flame is located 8 cm downstream of the dump plane, i.e. $L_f = 1.06$ m. Note that these values change with operating conditions, resulting in small differences in the resonant frequencies shown in Fig. 13. The total acoustic length of the combustor, $L_{tot} = 6.0$ m, is consistent with the observations in Fig. 11, where $p' \approx 0$ around 6 meters downstream of the choke plate.

The observed resonant frequencies as well as the modeshapes are predicted well using the acoustic model. The pressure nodes of

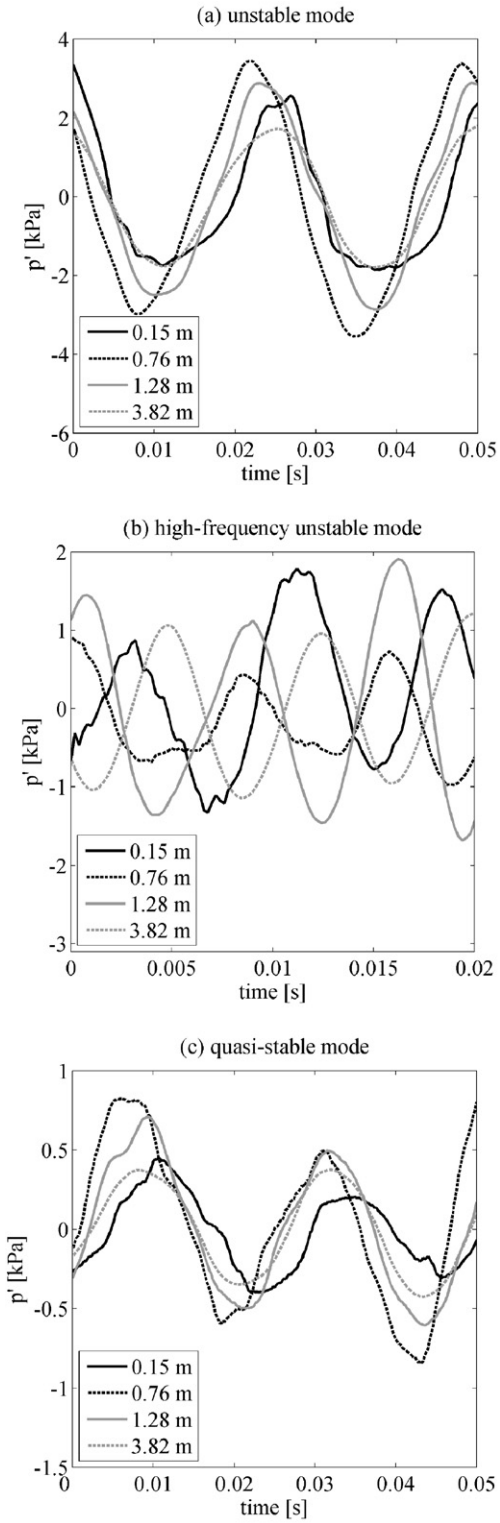

Fig. 10. Pressure oscillations measured at four different locations along the combustor length corresponding to the (a) unstable mode: $\phi = 0.80$, Re = 8500, $T_{in} = 300$ K, no hydrogen enrichment; (b) high-frequency unstable mode: $\phi = 0.70$, Re = 6500, $T_{in} = 500$ K, 50% by volume hydrogen enrichment; (c) quasi-stable mode: $\phi = 0.70$, Re = 8500, $T_{in} = 300$ K, no hydrogen enrichment.

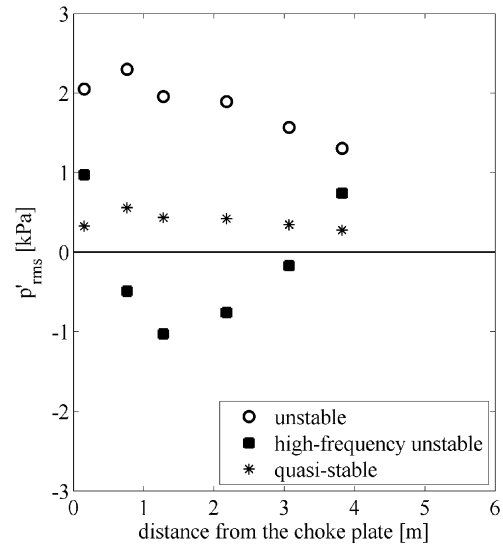

Fig. 11. Pressure modeshapes corresponding to the unstable, high-frequency unstable and quasi-stable operating modes.

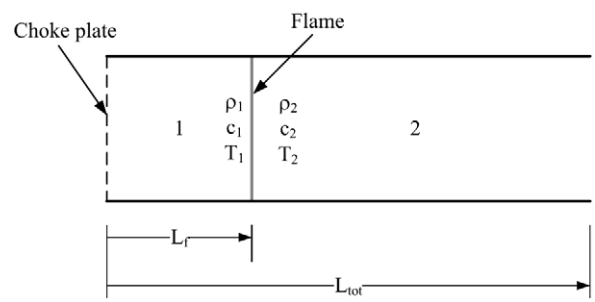

Fig. 12. Schematic diagram of the combustor used in the acoustic model.

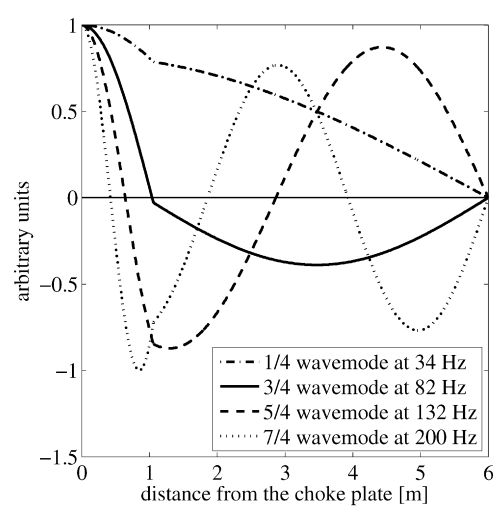

Fig. 13. Pressure modeshapes calculated from the acoustic model corresponding to the 1/4, 3/4, 5/4 and 7/4 wavemodes.

the 5/4 wavemode occur around 0.8 m and 2.9 m, which matches closely the experimental measurements shown in Fig. 11. These results clearly prove that the unstable and high-frequency unstable operating modes of the combustor correspond to the 1/4 and the 5/4 wavemodes, respectively. The resonant frequencies associated with different acoustic modes of the combustor correspond to distinct bands in Figs. 7 and 8. However the amplitude is significantly higher at the excited frequencies (the bands are much darker) corresponding to either the 1/4 or the 5/4 acoustic modes depending on the operating conditions. The figure also shows that the location of the flame is very close to a pressure node in the 3/4 wavemode, where the pressure amplitude is very low. This is the reason why the 3/4 wavemode was not excited.

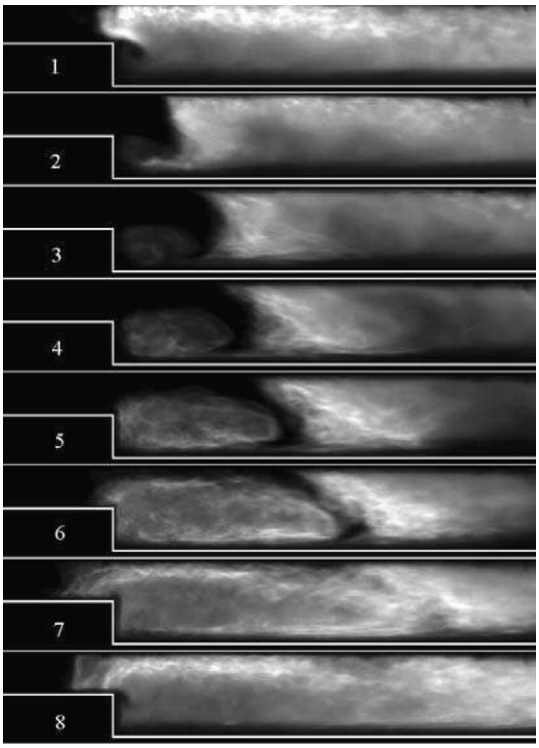

Fig. 14. Flame images of the unstable mode. $\phi = 0.80$, Re = 8500, $T_{in} = 300$ K, no hydrogen enrichment. The resonant frequency is 36 Hz. The time between the frames is 4 ms.

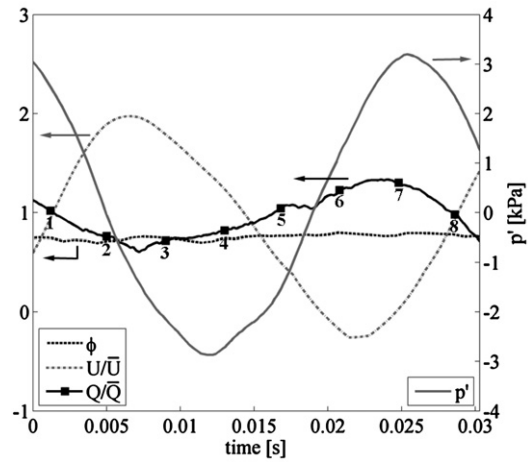

Fig. 15. Simultaneous heat release rate, velocity, pressure and equivalence ratio measurements corresponding to the unstable mode shown in Fig. 14.

3.3. Flame images

In order to distinguish between the dynamics at different operating modes, we examine flame images extracted from high speed videos of typical unstable, high-frequency unstable, quasi-stable and stable operating modes and describe corresponding unsteady pressure, velocity, heat release rate and equivalence ratio measurements. The pressure is measured using the sensor located 1.28 m downstream of the choke plate under all operating modes, which is close to the flame location.

3.3.1. Unstable mode

In Fig. 14, we show a sequence of flame images corresponding to one cycle of the unstable mode taken at $\phi = 0.80$, Re = 8500, $T_{in} = 300$ K, without hydrogen enrichment. Fig. 15 shows simultaneous heat release rate, velocity, pressure and equivalence ratio measurements covering the same time period as the images shown in Fig. 14. The resonant frequency is 36 Hz, which corresponds to the quarter wave mode of the combustor. Since we use a hot wire anemometer to perform the velocity measurements, we do not have information about the direction of the flow. Therefore, the sign of the hot wire reading is corrected to account for flow reversal. In the unstable mode, a wake vortex is formed in the recirculation zone downstream of the step (1). At this moment, the velocity is rising, and the acceleration and the pressure are at their maxima. As the velocity at the step increases, the vortex convects downstream, while moving toward the upper wall of the combustor. As a result, a packet of unburned reactants forms between the growing vortex and the flame of the previous cycle (2, 3). In addition, the heat release rate is decreasing following the end of the intense burning from the previous cycle. Between instants 2 and 3, heat release rate reaches its minimum, and the velocity is near its maximum value. As the velocity drops from its maximum value, the vortex continues to move downstream toward the upper wall of the combustor. The heat release rate starts to rise as the reactant packet sandwiched between the flame from the earlier cycle and the flame surrounding the new vortex begins to burn (4–6). As the velocity reaches its minimum value between images 6 and 7, the vortex reaches the upper wall of the combustor, and the reactant packet burns by two "advancing" flames on both sides causing intense burning. As a result, the heat release rate reaches its maximum near the moment of maximum pressure, resulting in the positive feedback between the unsteady heat release rate and the pressure, which drives the instability (7). The flame moving upstream of the step indicates very small or negative velocity at the step (6, 7). Next, as the velocity starts to rise again, the heat release rate starts to drop, fresh reactants enter the flame anchoring zone and the cycle repeats. Since the fuel injector is located far enough from the combustion zone, the equivalence ratio remains nearly constant throughout the cycle. The maximum pressure amplitude is near 3 kPa.

3.3.2. High-frequency unstable mode

When the inlet mixture is preheated to higher temperatures, we observe instability at higher frequencies. Fig. 16 shows a sequence of flame images corresponding to one cycle of the high-frequency unstable mode observed at $\phi = 0.70$, Re = 6500, $T_{in} = 500$ K, with 50% by volume hydrogen enrichment. Fig. 17 shows simultaneous heat release rate and pressure measurements covering the same time period as the images shown in Fig. 16. At elevated inlet temperatures we were not able to do point velocity measurements using the hot wire anemometer. Equivalence ratio measurements are not performed at this operating mode because the quality of the photodiode signal drops significantly as the inlet temperature increases. However, we expect the equivalence ratio to remain temporally constant since we placed the fuel injector far upstream of the step and as verified at $T_{in} = 300$ K. The resonant frequency is 132 Hz (which corresponds to the 5/4 wavemode of the combustor). The observed combustion dynamics are very similar to the unstable mode described before, except that the unsteady flame dynamics are faster in this case. In this operating mode, the vortex forms at the step when the pressure is at a minimum. Since the step is located between a pressure node and a pressure antinode (see Fig. 13), u' leads p' by 90 degrees. That instant again corresponds to the moment of maximum acceleration, similar to the unstable case. As a result of the higher frequency, the flame is more compact and the wake vortex reaches the upper wall of the combustor closer to the step. The heat release rate and pressure oscillations are nearly in phase, resulting in the observed instability. The more compact flame causes smaller variations in the flame surface area, resulting in smaller amplitude

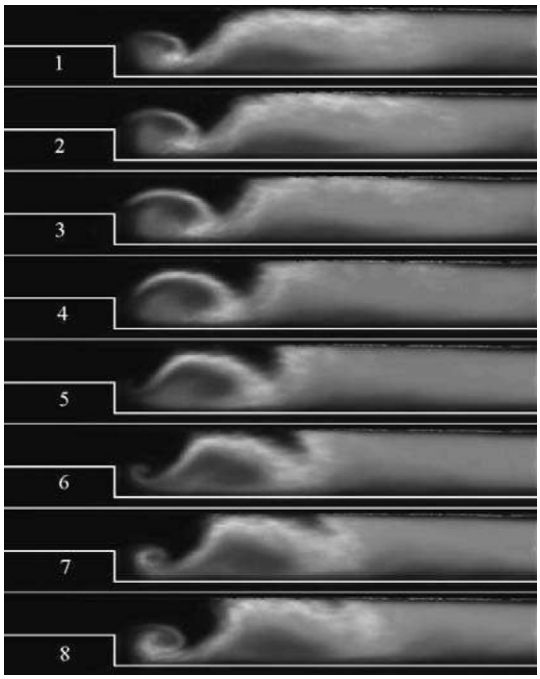

Fig. 16. Flame images of the high-frequency unstable mode. $\phi = 0.70$, Re = 6500, $T_{in} = 500$ K, with 50% by volume hydrogen enrichment. The resonant frequency is 132 Hz. The time between the frames is 1.1 ms.

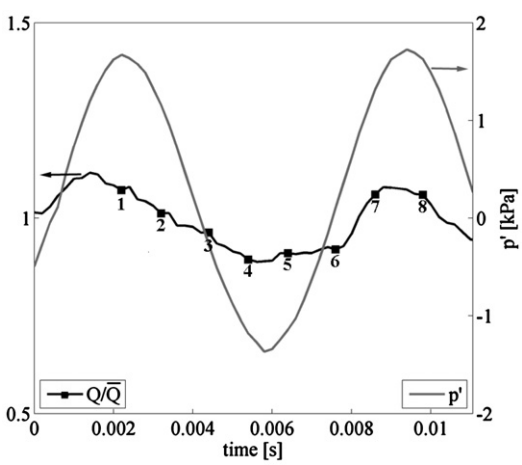

Fig. 17. Simultaneous heat release rate and pressure measurements corresponding to the high-frequency unstable mode shown in Fig. 16.

heat release rate oscillations compared to the unstable case. The unsteady pressure amplitude is also lower than the unstable case as a result of the smaller amount of energy supplied to the acoustic field due to heat release. The maximum pressure amplitude is around 1.5 kPa.

3.3.3. Quasi-stable mode

When the equivalence ratio is reduced while the combustor is operating in the unstable mode, the noise from the combustor decreases significantly as the equivalence ratio reaches a certain threshold, which depends on the operating conditions. This is the quasi-stable operating mode of the combustor which corresponds to the 140–150 dB plateau observed in Fig. 9 at all operating conditions.

In Fig. 18, we show a sequence of flame images corresponding to the quasi-stable mode observed at $\phi = 0.70$, Re = 8500, $T_{in} = 300$ K, without hydrogen enrichment. Fig. 19 shows simultaneous heat release rate, velocity, pressure and equivalence ratio

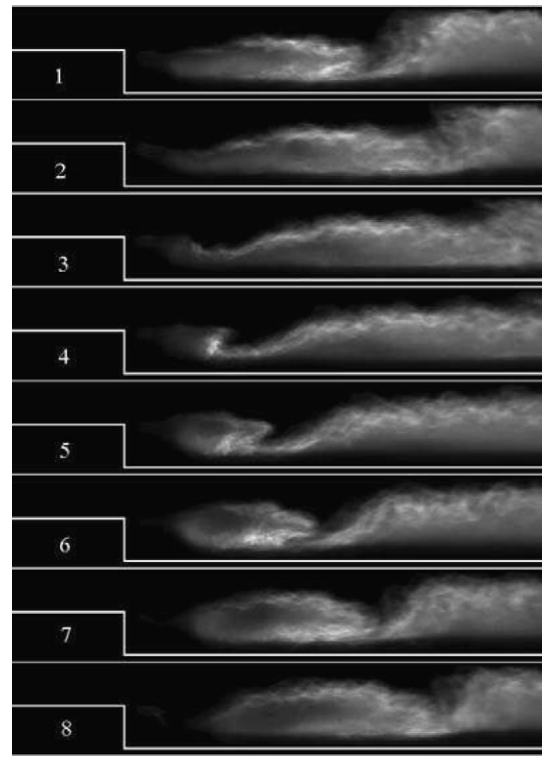

Fig. 18. Flame images of the quasi-stable mode. $\phi = 0.70$, Re = 8500, $T_{in} = 300$ K, without hydrogen enrichment. The resonant frequency is 38 Hz. The time between the frames is 3.8 ms.

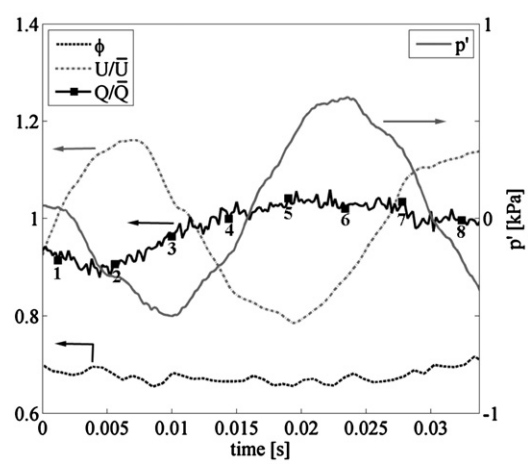

Fig. 19. Simultaneous heat release rate, velocity, pressure and equivalence ratio measurements corresponding to the quasi-stable mode shown in Fig. 18.

measurements covering the same time period as the images shown in Fig. 18. The resonant frequency is 38 Hz, again corresponding to the 1/4 wavemode of the combustor. Similar to the unstable mode operation, in the quasi-stable case, a vortex is formed at the edge of the step near the moment of maximum acceleration (1). The vortex grows and convects downstream, forming a packet of reactants (4–8). Unlike the unstable case, the vortex never grows to reach the top wall, and the reactant packet is not pinched off by the vortex. The burning is concentrated near the reactant pocket. Compared to the unstable case, the burning of the pocket is not sudden and produces relatively weak heat release rate fluctuations as shown in Fig. 19. This results in significantly lower sound pressure levels compared to the unstable mode. The maximum pressure amplitude is 0.6 kPa.

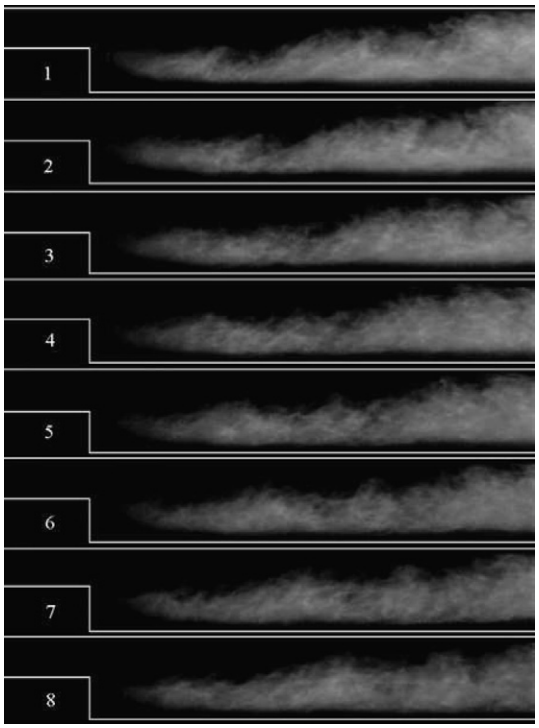

Fig. 20. Flame images of the stable mode. $\phi = 0.57$, Re = 6500, $T_{in} = 300$ K, with 50% by volume hydrogen enrichment. The time between the frames is 4 ms.

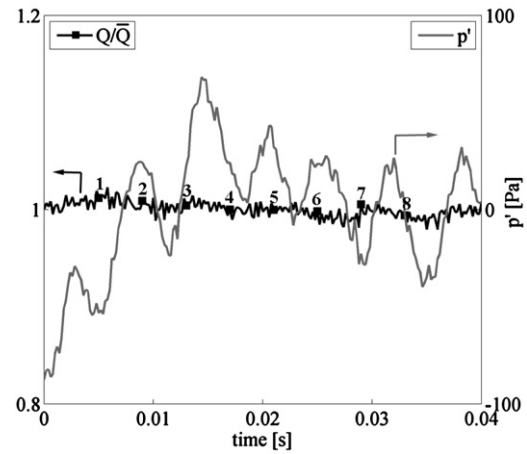

Fig. 21. Simultaneous heat release rate and pressure measurements corresponding to the stable mode shown in Fig. 20.

3.3.4. Stable mode

Very close to the lean blowout limit, the combustor operates in the stable mode, resulting in nearly silent burning. As the inlet temperature increases, the stable mode becomes narrower as shown in Fig. 9. In Fig. 20, we show a sequence of flame images corresponding to the stable mode observed at $\phi = 0.57$, Re = 6500, $T_{in} = 300$ K, with 50% by volume hydrogen enrichment. Fig. 21 shows simultaneous heat release rate and pressure measurements covering the same time period as the images shown in Fig. 20. In the stable case, small vortices are shed from the step at high frequencies. The vortex shedding frequency does not correspond to one of the low frequency acoustic modes of the combustor. Thus, there is no coupling between the heat release rate and pressure as shown in Fig. 21. The heat release rate is steady, and the pressure oscillates at very small amplitude and high frequency, which corresponds to the vortex shedding frequency.

3.4. Theory

In this section we develop a theory to predict the conditions at which the sudden transition between different dynamics operating modes take place. The high speed flame images of the unstable and high-frequency unstable operating modes revealed that during an unstable cycle, the flame is convoluted around a vortex forming in the wake of the step. Moreover, careful inspection of the results in Figs. 14 and 16 shows that as the wake vortex reaches the upper wall of the combustor, the reactant packet formed between adjacent flames burns, causing the heat release rate to reach its maximum value. For both operating modes, the vortex forms at the moment of maximum acceleration. We define the time elapsed between the moment of vortex formation and the moment when the vortex reaches the upper wall (moment of maximum heat release rate) as τ_{vor}. We assume that the vortex moves toward the wall predominantly as a result of the expansion velocity in the vertical direction. The expansion velocity is proportional to the heat release rate parameter $S_T(\rho_u/\rho_b - 1)$ where S_T is the turbulent burning velocity and ρ_u and ρ_b are the unburned and burned mixture densities [35]. Assuming a linear relationship between the turbulent and laminar burning velocities, the vertical velocity of the vortex, v_{vor}, is:

$$v_{vor} = \frac{S_L}{c}(\rho_u/\rho_b - 1) \tag{11}$$

where S_L is the laminar burning velocity of the mixture. The constant of proportionality between the expansion velocity and the heat release rate parameter, and the proportionality between the turbulent and laminar burning velocities are represented by an arbitrary constant c. Representing the vertical distance between the corner of the step where the vortex forms and the upper wall as H, we obtain $\tau_{vor} = H/v_{vor}$. Substituting v_{vor} from Eq. (11):

$$\tau_{vor} = \frac{cH}{S_L(\rho_u/\rho_b - 1)}. \tag{12}$$

In Fig. 22, we schematically show τ_{vor}, and the time between the maximum heat release rate and maximum pressure, τ_{qp}, for the unstable and high-frequency unstable cases. The quasi-stable to unstable, and unstable to high-frequency unstable transitions occur when the pressure and heat release rate oscillations become in phase, that is $\tau_{pq} = NT$, where N is an integer and T is the oscillation period. According to Fig. 22, the quasi-stable to unstable transition takes place when (for $N = 1$)

$$\tau_{vor} = \frac{1}{f_{1/4}} \tag{13}$$

and the unstable to high-frequency unstable transition takes place when

$$\tau_{vor} = \frac{3}{2}\frac{1}{f_{5/4}} \tag{14}$$

where $f_{1/4}$ and $f_{5/4}$ are the resonant frequencies corresponding to unstable and high-frequency unstable modes. Using Eqs. (12) and (13), for the quasi-stable to unstable transition (qs → u):

$$S_L(\rho_u/\rho_b - 1)|_{qs\to u} = cHf_{1/4}. \tag{15}$$

Similarly, using Eqs. (12) and (14), for the unstable to high-frequency unstable transition (u → hfu):

$$S_L(\rho_u/\rho_b - 1)|_{u\to hfu} = \frac{2}{3}cHf_{5/4}. \tag{16}$$

The left-hand sides of Eqs. (15) and (16), i.e. the heat release parameters at transition, depend on the operating conditions (ϕ, fuel composition and T_{in}). For a set of operating conditions satisfying

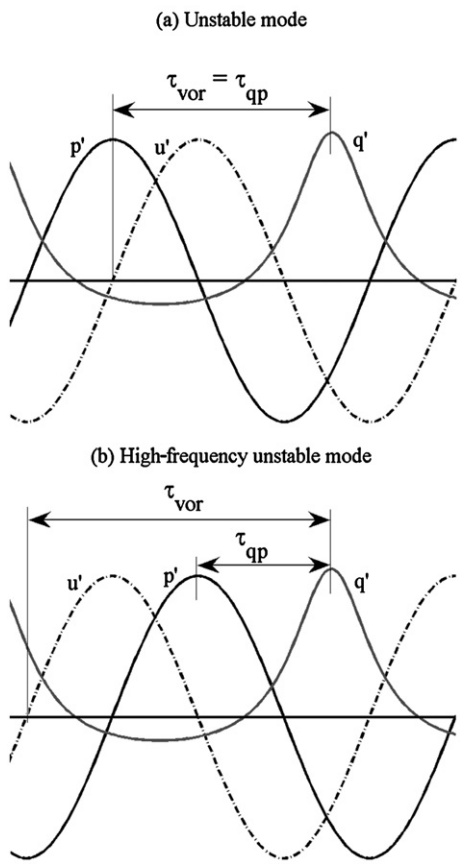

Fig. 22. Schematic diagram showing pressure, velocity and heat release rate oscillations in a typical cycle corresponding to the (a) unstable; (b) high-frequency unstable operating modes.

Eq. (15) or (16), we expect quasi-stable to unstable, or unstable to high-frequency unstable transitions to take place, respectively.

In order to test this hypothesis, we regenerate the response curves of the combustor—the OASPL as function of the equivalence ratio—shown in Fig. 9 by plotting the OASPL against the heat release parameter. Since, the dependence on all the operating conditions are represented by this parameter alone, the response curves should collapse onto a global curve independent of the operating conditions. However, attempting to collapse the data using the laminar burning velocity did not produce satisfactory results. For this reason, we focused our attention in modifying the laminar burning velocity by including the role of strain on laminar flames over the same range of equivalence ratios, fuel compositions and inlet temperatures. Especially with different fuel compositions, strain significantly impacts the burning velocity [36].

We investigated the effect of strain by simulating a laminar flame stabilized in a planar stagnation flow. Reactants are supplied on one side, the equilibrium state products of the reactants are supplied on the other, and the flame stabilizes in the vicinity of the stagnation point. The corresponding potential flow velocity field is characterized by the time-varying strain rate parameter a. The one-dimensional governing equations for the flame structure are found by using a boundary layer approximation across the flame thickness. The governing equations are discretized using an implicit finite-difference method. The solution is obtained using a preconditioned inexact Newton–Krylov method. Cantera [37] is used to evaluate the chemical source terms and the various physical properties. Because of its strong influence on the diffusion of hydrogen, the Soret contribution to diffusive fluxes is included. Details of the numerical method may be found in previous papers [38,39]. We

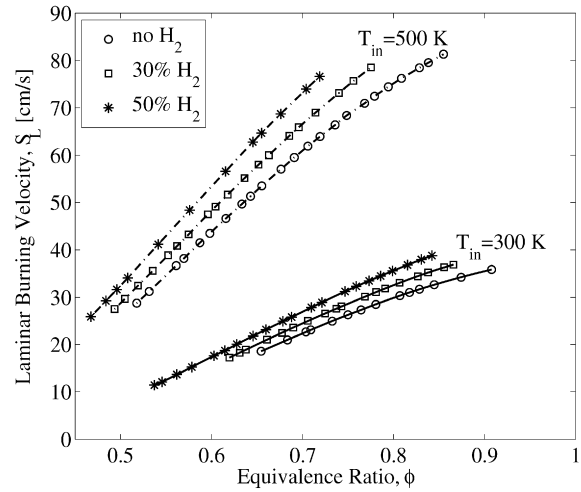

Fig. 23. Laminar burning velocities of propane–hydrogen flames as a function of equivalence ratio, fuel composition and inlet temperature.

use a reduced chemical kinetic-mechanism for propane combustion developed at the UCSD Center of Energy Research [40].

There are many possible ways to define a flame speed for a strained flame. Here, we use the consumption speed, S_c, found by integrating the energy equation across the flame. The consumption speed is defined as

$$S_c = \frac{\int_{-\infty}^{\infty} q'''/c_p\,dy}{\rho_u(T_b - T_u)} \tag{17}$$

where q''' is the volumetric heat release rate, c_p is the specific heat of the mixture, y is the coordinate normal to the flame, ρ_u is the unburned mixture density, and T_u and T_b are the unburned and burned temperature, respectively. As the strain rate parameter a approaches 0, the consumption speed approaches the laminar burning velocity.

To understand the effects of strain on propane–hydrogen flames, we ran a comprehensive set of numerical simulations covering the fuel compositions, equivalence ratios and inlet temperatures used in Fig. 9, and strain rates ranging from 10 to 6000 s^{-1}. In all cases, the oxidizer was air and the pressure was atmospheric.

The laminar burning velocity, i.e. at $a = 0$, as a function of equivalence ratio for each fuel composition considered in the experiment and inlet temperatures of 300 K and 500 K is shown in Fig. 23. The laminar burning velocity is determined by extrapolating the consumption speed to a strain rate of zero, analogous to the commonly employed technique for determining laminar burning velocity from experimentally observed strained flames. The burning velocity rises substantially with inlet temperature as expected. For the equivalence ratios considered here, increasing hydrogen concentration raises the burning velocity despite the negligible change in the adiabatic flame temperature as shown in Fig. 24. Clearly, the impact of hydrogen enrichment is physical and not chemical.

Fig. 25 shows the effect of strain rate on flames with equivalence ratio $\phi = 0.70$ for each of the fuel compositions considered in the experiment, and inlet temperatures of 300 K and 500 K. Asymptotic analysis of strained flames predicts that the consumption speeds of flames with Lewis numbers greater than unity decrease with positive strain, and that decrease is proportional to the departure of the Lewis number from unity. The results shown in this figure are consistent with the expected trend; the consumption speed of the flames with highest hydrogen content (Lewis number is closest to unity) has the weakest response to increasing the strain rate. When the inlet temperature is raised, the response

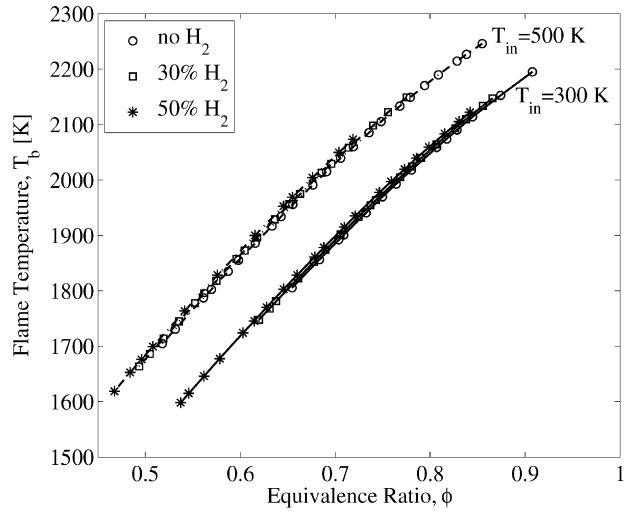

Fig. 24. Adiabatic flame temperatures of propane–hydrogen flames as a function of equivalence ratio, fuel composition and inlet temperature.

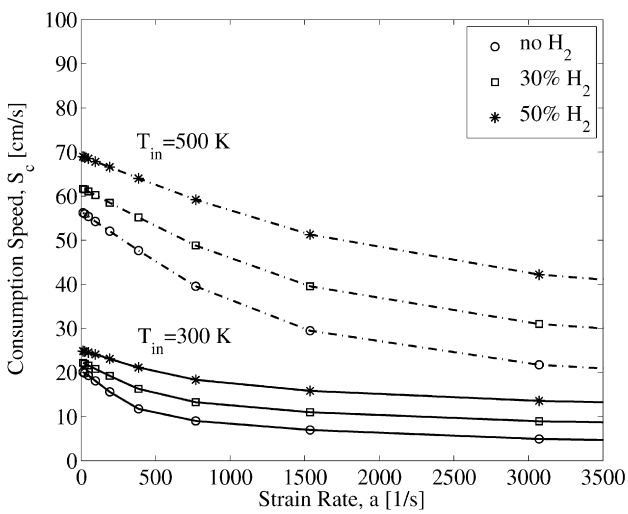

Fig. 25. Effect of strain rate on consumption speed of flames at $\phi = 0.70$ with varying fuel compositions and inlet temperature.

on the strain rate becomes stronger for all fuel compositions. (In the case of syngas flames, we observed opposite trends for the response of flames with different hydrogen content to the strain rate, since in this case the Lewis number is less than unity. Increasing the amount of hydrogen in syngas flames increases the departure of Lewis number from unity, resulting in stronger response of the consumption speed to strain rate [36,41].)

In Fig. 26, we plot the consumption speed as a function of the equivalence ratio for each fuel composition, and inlet temperatures of 300 K and 500 K, at a strain rate of $a = 2000\ \mathrm{s}^{-1}$. The dependence of the consumption speed on the equivalence ratio is nearly linear for each fuel composition and inlet temperature. When we compare this figure to the unstrained case, Fig. 23, we observe that the strained flames with less hydrogen content are more strongly impacted by the strain rate when compared to higher hydrogen content flames, especially at high equivalence ratios. The Lewis number dependent response of the consumption speed to stretch is greater at higher equivalence ratios.

These results clearly demonstrate that the strained flame consumption speed S_c is more sensitive to the operating conditions than the laminar burning velocity. When using the consumption to reduce the experimental data, there is one undetermined con-

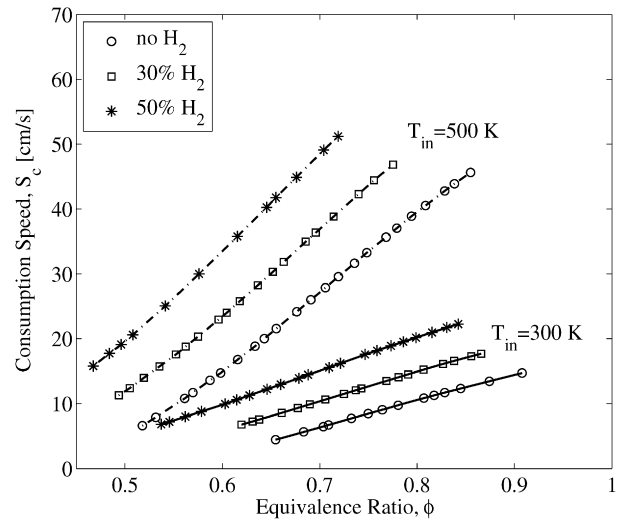

Fig. 26. Consumption speed of propane–hydrogen flames at a fixed strain rate of $a = 2000\ \mathrm{s}^{-1}$ as a function of equivalence ratio at different fuel compositions and inlet temperatures.

stant a, which can be thought of as a typical strain rate averaged over the entire flame. Here, we select values of a which produces the best correspondence with the experimental data at each inlet temperature, i.e. the value for which the OASPL as a function the heat release parameter looks essentially the same for all fuel compositions at each inlet temperature. In Fig. 27 we plot the OASPL against the heat release parameter using the consumption speed for the burning velocity for the data in Fig. 9 using strain rates of $2000\ \mathrm{s}^{-1}$, $2300\ \mathrm{s}^{-1}$, $2700\ \mathrm{s}^{-1}$ and $3000\ \mathrm{s}^{-1}$ for inlet temperatures of 300 K, 400 K, 500 K and 600 K, respectively. As the inlet temperature increases, the average inlet velocity also increases for the same Reynolds number. Since the strain rate is proportional to the velocity, the strain rates are higher at higher inlet temperatures, consistent with the values used in generating Fig. 27. Moreover, as the inlet temperature increases, the average flame location becomes closer to the step, where the strain rate is the highest. Here we see that, for the most part, the data are well-correlated with the heat release parameter at these values of the strain rates.

We next proceed to further validate the collapse of the data when plotted against the heat release parameter. We plot the OASPL against the heat release parameter in Fig. 28 selecting arbitrary fuel compositions and inlet temperatures from Fig. 27. The figure shows that the transitions between different operating modes—independent of the fuel composition, equivalence ratio and the inlet temperature—are correlated well with the heat release parameter, which is a global parameter representing the flame–vortex interaction dynamics. As mentioned before, since the combustor shows strong dependence on the history of the operating conditions, minor differences at different operating conditions are expected. The transition from the quasi-stable to unstable mode occurs at $S_c(\rho_u/\rho_b - 1) \approx 50$ cm/s, and the unstable mode to high-frequency unstable mode transition takes place at $S_c(\rho_u/\rho_b - 1) \approx 120$ cm/s.

Finally, we verify the theory derived in Eqs. (15) and (16). We define r as the ratio between the heat release parameters at the unstable to high-frequency unstable transition and at the quasi-stable to unstable transition. From Eqs. (15) and (16):

$$r = \frac{S_L(\rho_u/\rho_b - 1)|_{\mathrm{u}\to\mathrm{hfu}}}{S_L(\rho_u/\rho_b - 1)|_{\mathrm{qs}\to\mathrm{u}}} = \frac{2}{3}\frac{f_{5/4}}{f_{1/4}}. \tag{18}$$

The resonant frequencies corresponding to high-frequency unstable and unstable cases, respectively, are 132 Hz and 36 Hz (see Section 3.3). Thus, the ratio predicted by the theory is: $r = 2.44$. The

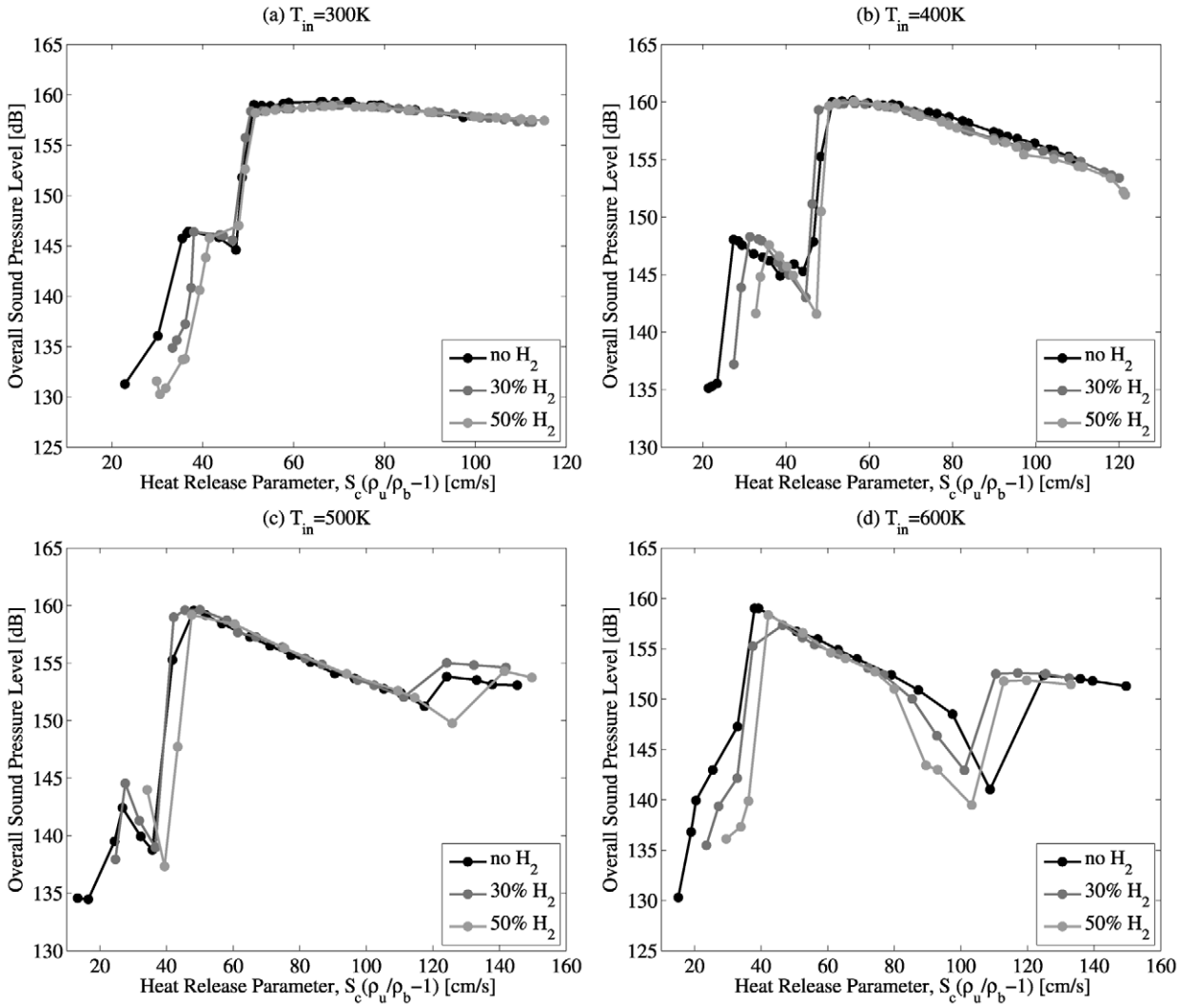

Fig. 27. OASPL as a function of heat release parameter, $S_c(\rho_u/\rho_b - 1)$, at different hydrogen concentrations for (a) $T_{in} = 300$ K, $a = 2000$ s^{-1}; (b) $T_{in} = 400$ K, $a = 2300$ s^{-1}; (c) $T_{in} = 500$ K, $a = 2700$ s^{-1}; (d) $T_{in} = 600$ K, $a = 3000$ s^{-1}.

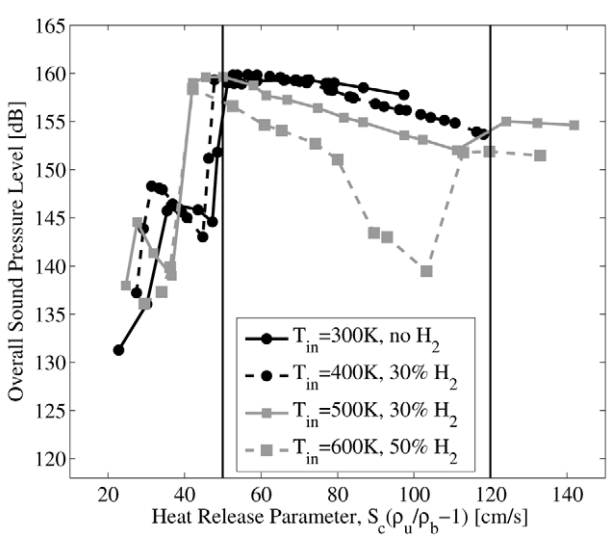

Fig. 28. OASPL as a function of heat release parameter, $S_c(\rho_u/\rho_b - 1)$, at different hydrogen concentrations and inlet temperatures.

ratio found from the experimental measurements is: $r = 120/50 = 2.40$. The theory predicts the experimental measurements with great accuracy! The constant c in Eqs. (15) and (16) is calculated using the experimental heat release rate parameter at the quasi-stable to unstable transition:

$$c = \frac{S_L(\rho_u/\rho_b - 1)|_{\mathrm{qs}\to\mathrm{u}}}{Hf_{1/4}} \tag{19}$$

which is 0.7.

This analysis show that the stability of the combustor is governed by the transport of the large wake vortex to the upper wall of the combustor. The complicated flame–vortex interaction dynamics can be reduced to a single parameter, $S_c(\rho_u/\rho_b - 1)$, which carries information about the interaction between the flame and the hydrodynamics with different reactant mixture compositions and the volume expansion as a result of the chemical reaction. The former is represented by the consumption speed S_c, and the latter is represented by $\rho_u/\rho_b - 1$.

4. Conclusions

In this paper, we systematically investigated the stability map and combustion dynamics in a backward-facing step combustor

where the dynamics are driven solely by flame–vortex interactions, that is in the absence of equivalence ratio oscillations. Simultaneous pressure, velocity, heat release rate and equivalence ratio measurements and high-speed video from the experiments were used to identify and characterize several distinct operating modes. We observed four distinct operating modes of the combustor depending on the inlet temperature and the equivalence ratio. At high but lean equivalence ratios, the combustor is unstable as a result of the strong interaction between the large unsteady wake vortex and the flame. The flame propagates upstream of the step during a part of the cycle. At intermediate equivalence ratios, the combustor is quasi-stable with the flame staying attached to the step at all times. Near the blowout limit, long stable flames are observed. When the inlet temperature is atmospheric, the quasi-stable and the unstable flames couple with the 1/4 wavemode of the combustor. When the inlet temperature is increased, at high but lean equivalence ratios, the 5/4 wavemode of the combustor is excited, increasing the resonant frequency significantly. This operating mode is the high-frequency unstable mode, where the flame is more compact compared to the unstable case. Increasing the inlet temperature and hydrogen concentration shifts the response curves of the combustor toward lower equivalence ratios. As the inlet temperature is increased, the stable operating band becomes narrower. The vertical velocity of the large wake vortex is proportional to the heat release parameter, and controls the moment when the vortex reaches the upper wall of the combustor producing a heat release pulse. We demonstrated that the consumption speed of strained flames can be used to collapse the response curves describing the transitions among different dynamic modes onto a function of the heat release parameter alone. We predict the critical values of the heat release parameter at which transitions between quasi-stable to unstable, and unstable to high-frequency unstable operating modes take place, which may provide significant benefits during the design stage of the combustors.

In future studies, we plan to extend this work by performing high speed PIV measurements in order to determine the instantaneous velocity fields and to define the origin of the strain rates used to calculate the flame consumption speeds. We will move the fuel injector closer to the step to introduce temporal equivalence ratio variations and analyze their impact on the flame–vortex interaction mechanism.

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